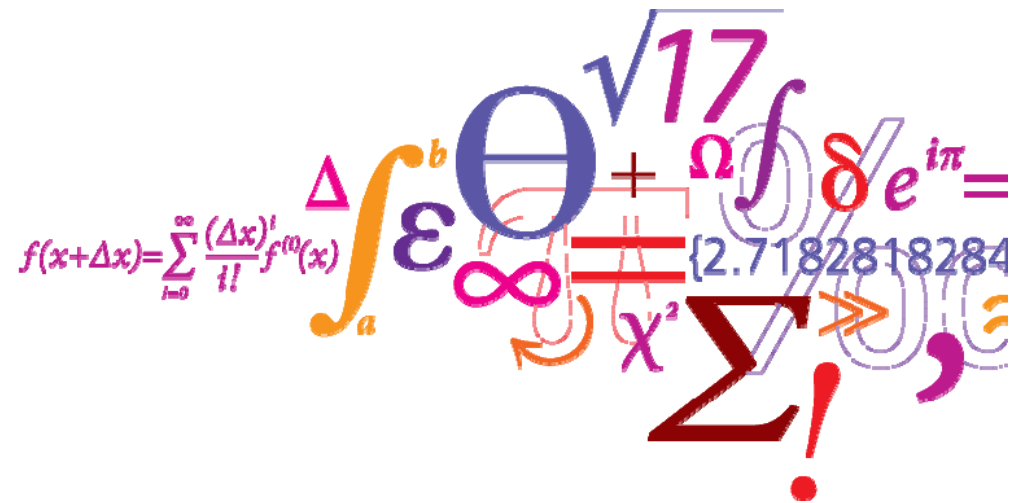


31792 -- Advanced Optimization and Game Theory for Energy Systems

Lecture 2: Market clearing as an equilibrium problem

Jalal Kazempour

January 5, 2021



Recap

Market clearing: a simple example

An extended example: two generators (G1 and G2) and two elastic demands (D1 and D2)



Capacity: 100 MW
Offer price: \$12/MWh



Capacity: 80 MW
Offer price: \$20/MWh

Demand D1

Maximum load: 100 MW
Bid price: \$40/MWh

Demand D2

Maximum load: 50 MW
Bid price: \$35/MWh

Market outcomes:

- Productions of G1 and G2: **100 MW and 50 MW**
- Consumptions of D1 and D2: **100 MW and 50 MW**
- Market-clearing price: **[20-35] \$/MWh**

Market clearing as an **optimization** problem



$$\underset{p^{G1}, p^{G2}, p^{D1}, p^{D2}}{\text{Maximize}} \quad SW = [40p^{D1} + 35p^{D2}] - [12p^{G1} + 20p^{G2}] \quad (1a)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad (1b)$$

$$0 \leq p^{D2} \leq 50 \quad (1c)$$

$$0 \leq p^{G1} \leq 100 \quad (1d)$$

$$0 \leq p^{G2} \leq 80 \quad (1e)$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0 \quad : \lambda \quad (1f)$$

Market clearing as an optimization problem

$$\underset{p^{G1}, p^{G2}, p^{D1}, p^{D2}}{\text{Maximize}} \quad SW = \underbrace{[40p^{D1} + 35p^{D2}]}_{\text{Utility of demands}} - \underbrace{[12p^{G1} + 20p^{G2}]}_{\text{Cost of generators}} \quad (1a)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{Consumption limits} \quad (1b)$$

$$0 \leq p^{D2} \leq 50 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \quad (1c)$$

$$0 \leq p^{G1} \leq 100 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{Generation limits} \quad (1d)$$

$$0 \leq p^{G2} \leq 80 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \quad (1e)$$

$$\underbrace{p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0}_{\text{Power balance equality}} \quad \left(: \lambda \right) \quad (1f)$$

Dual variable: market-clearing price

Discussion



Question:

How to make sure all market participants (i.e., G1, G2, D1 and D2) are satisfied with the market-clearing outcome, and would not deviate from it?

Discussion



Question:

How to make sure all market participants (i.e., G1, G2, D1 and D2) are satisfied with the market-clearing outcome, and would not deviate from it?

Let's develop an optimization problem for each market participant!

Discussion

Question:

How to make sure all market participants (i.e., G1, G2, D1 and D2) are satisfied with the market-clearing outcome, and would not deviate from it?

Let's develop an optimization problem for each market participant!

Question:

What is the objective of each generator?

What is the objective of each elastic demand?

Discussion

Question:

How to make sure all market participants (i.e., G1, G2, D1 and D2) are satisfied with the market-clearing outcome, and would not deviate from it?

Let's develop an optimization problem for each market participant!

Question:

What is the objective of each generator? **Profit maximization!**

What is the objective of each elastic demand? **Utility maximization!**

Discussion

Question:

How to make sure all market participants (i.e., G1, G2, D1 and D2) are satisfied with the market-clearing outcome, and would not deviate from it?

Let's develop an optimization problem for each market participant!

Question:

What is the objective of each generator? **Profit maximization!**

What is the objective of each elastic demand? **Utility maximization!**

Question:

How to calculate a generator's profit?

How to calculate an elastic demand's utility?

Discussion

Question:

How to make sure all market participants (i.e., G1, G2, D1 and D2) are satisfied with the market-clearing outcome, and would not deviate from it?

Let's develop an optimization problem for each market participant!

Question:

What is the objective of each generator? **Profit maximization!**

What is the objective of each elastic demand? **Utility maximization!**

Question:

How to calculate a generator's profit? **Production level x [market price – production cost]**

How to calculate an elastic demand's utility? **Consumption level x [bid price – market price]**

Optimization problem for each market player



Optimization problem for each market player



For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

Optimization problem for each market player



For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

Optimization problem for each market player



For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

Optimization problem for each market player



For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Optimization problem for each market player



For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Market price λ is a given value (treated as a parameter) within each optimization problem!

Optimization problem for each market player

For generator G1:

$$\begin{aligned} &\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12) \\ &\text{subject to:} \\ &0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1} \end{aligned}$$

For generator G2:

$$\begin{aligned} &\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20) \\ &\text{subject to:} \\ &0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2} \end{aligned}$$

For demand D1:

$$\begin{aligned} &\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda) \\ &\text{subject to:} \\ &0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1} \end{aligned}$$

For demand D2:

$$\begin{aligned} &\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda) \\ &\text{subject to:} \\ &0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2} \end{aligned}$$

Market price λ is a given value (treated as a parameter) within each optimization problem!

Question:

How do market players contribute to market price formation? Do we need an extra condition?

Optimization problem for each market player

For generator G1:

$$\begin{aligned} &\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12) \\ &\text{subject to:} \\ &0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1} \end{aligned}$$

For generator G2:

$$\begin{aligned} &\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20) \\ &\text{subject to:} \\ &0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2} \end{aligned}$$

For demand D1:

$$\begin{aligned} &\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda) \\ &\text{subject to:} \\ &0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1} \end{aligned}$$

For demand D2:

$$\begin{aligned} &\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda) \\ &\text{subject to:} \\ &0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2} \end{aligned}$$

Market price λ is a given value (treated as a parameter) within each optimization problem!

Question:

How do market players contribute to market price formation? Do we need an extra condition? **Yes!**

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Power balance equality:

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0 \quad : \lambda$$

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Power balance equality:

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0 \quad : \lambda$$

Question: From mathematical perspective, is the above power balance equality “equivalent” to the following “unconstrained” optimization problem? Why?

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Power balance equality:

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0 \quad : \lambda$$

Assume this unconstrained optimization problem is being solved by a fictitious player, the so-called “**price-setter**”, who determines the market-clearing price by penalizing the power mismatch!

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$



Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

Question: Can we solve optimization problems above separately?

Optimization problem for each market player



For generator G1:

$$\begin{aligned} &\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12) \\ &\text{subject to:} \\ &0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1} \end{aligned}$$

For generator G2:

$$\begin{aligned} &\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20) \\ &\text{subject to:} \\ &0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2} \end{aligned}$$

For demand D1:

$$\begin{aligned} &\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda) \\ &\text{subject to:} \\ &0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1} \end{aligned}$$

For demand D2:

$$\begin{aligned} &\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda) \\ &\text{subject to:} \\ &0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2} \end{aligned}$$

Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

Question: Can we solve optimization problems above separately? **No! Why?**

Optimization problem for each market player

For generator G1:

$$\begin{aligned} & \underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12) \\ & \text{subject to:} \\ & 0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1} \end{aligned}$$

For generator G2:

$$\begin{aligned} & \underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20) \\ & \text{subject to:} \\ & 0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2} \end{aligned}$$

For demand D1:

$$\begin{aligned} & \underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda) \\ & \text{subject to:} \\ & 0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1} \end{aligned}$$

For demand D2:

$$\begin{aligned} & \underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda) \\ & \text{subject to:} \\ & 0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2} \end{aligned}$$

Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

- Market-clearing price is a variable for the price-setter, but a parameter for G1, G2, D1 and D2.
- Productions/consumptions are variables for G1, G2, D1 and D2, but parameters for the price-setter.

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

All five optimization problems above are linked, and should be solved all together!

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

This is a game-theoretic problem.

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

This is a game-theoretic problem.

This specific problem is also known as “**competitive equilibrium**” problem!

Optimization problem for each market player

For generator G1:

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

For generator G2:

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

For demand D1:

$$\underset{p^{D1}}{\text{Maximize}} \quad p^{D1}(40 - \lambda)$$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

For demand D2:

$$\underset{p^{D2}}{\text{Maximize}} \quad p^{D2}(35 - \lambda)$$

subject to:

$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Price-setter:

$$\underset{\lambda}{\text{Minimize}} \quad \lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$$

Discussion:

What kind of game-theoretic problem is it? Is it a “**non-cooperative**” game? Or a “**cooperative**” one?

Some seminal works on competitive equilibrium



MATHEMATICAL METHODS OF ORGANIZING AND PLANNING PRODUCTION*†

L. V. KANTOROVICH

Leningrad State University

1939

Contents

L. V. Kantorovich, "Mathematical methods of organizing and planning production," *Management Science*, vol. 6, no. 4, pp. 366–422, 1960.

SPATIAL PRICE EQUILIBRIUM AND LINEAR PROGRAMMING

By PAUL A. SAMUELSON*

I.—Introduction

Increasingly, modern economic theorists are going beyond the formulation of equilibrium in terms of such marginal equalities as marginal revenue equal to marginal costs or wage rate equal to marginal value product. Instead they are reverting to an earlier and more fundamental aspect of a maximum position: namely, that from the top of a hill, whether or not it is locally flat, all movements are downward.

P. A. Samuelson, "Spatial price equilibrium and linear programming," *American Economic Review*, vol. 42, no. 3, pp. 283–303, 1952.

ECONOMETRICA

VOLUME 22

July, 1954

NUMBER 3

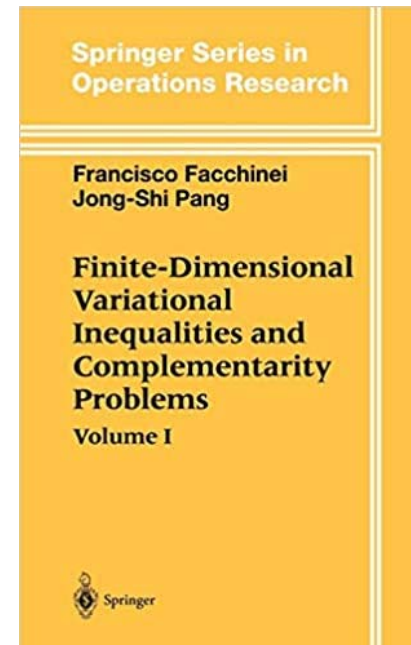
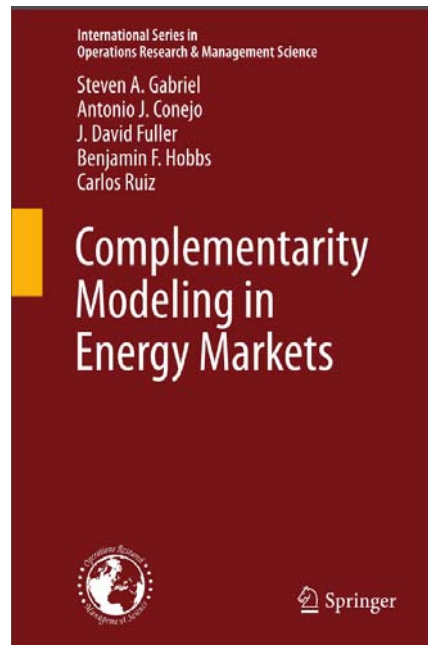
EXISTENCE OF AN EQUILIBRIUM FOR A COMPETITIVE ECONOMY

By KENNETH J. ARROW AND GERARD DEBREU¹

A. Wald has presented a model of production and a model of exchange and proofs of the existence of an equilibrium for each of them. Here proofs of the existence of an equilibrium are given for an *integrated* model of production, exchange and consumption. In addition the assumptions made on the technologies of producers and the tastes of consumers are significantly weaker than Wald's. Finally a simplification of the structure of the proofs has been made possible through use of the concept of an abstract economy, a generalization of that of a game.

K. J. Arrow and G. Debreu, "Existence of an equilibrium for a competitive economy," *Econometrica*, vol. 22, no. 3, pp. 265–290, 1954.

Relevant books and courses!



- Prof. Steven Gabriel's yearly short course at NTNU, "**Introduction Course in Complementarity Models and Equilibrium**": <https://www.ntnu.edu/studies/courses/l%C3%988806#tab=omEmnet>
- Prof. Uday Shanbhag's invited 5-day course at DTU in 2019. All video lectures are publicly available here: https://www.youtube.com/watch?v=PYXlzmW53k&list=PLKLR7D59yU0fuZTH5wjgov31D3DXta_l-

Market clearing as an **equilibrium** problem

- If a solution to our equilibrium problem exists, it will be a “**Nash equilibrium point**”, i.e.,



John Nash

No market participant can increase its profit by deviating unilaterally from the equilibrium solution!

Market clearing as an **equilibrium** problem

- If a solution to our equilibrium problem exists, it will be a “**Nash equilibrium point**”, i.e.,



John Nash

No market participant can increase its profit by deviating unilaterally from the equilibrium solution!

Discussion:

- Is our market-clearing problem a “Nash equilibrium” (NE) problem?
- Or, is it a “generalized” Nash equilibrium (GNE)?
- What is the difference of NE and GNE? Which one is more appealing?

Market clearing as an **equilibrium** problem

- If a solution to our equilibrium problem exists, it will be a “**Nash equilibrium point**”, i.e.,



John Nash

No market participant can increase its profit by deviating unilaterally from the equilibrium solution!

Recall the first question:

How to make sure all market participants are satisfied with the outcome of market-clearing **optimization** problem, and would not deviate from it?

Market clearing as an **equilibrium** problem

- If a solution to our equilibrium problem exists, it will be a “**Nash equilibrium point**”, i.e.,



John Nash

No market participant can increase its profit by deviating unilaterally from the equilibrium solution!

Recall the first question:

How to make sure all market participants are satisfied with the outcome of market-clearing **optimization** problem, and would not deviate from it?

And new questions:

- Which problem should we solve to clear the market (optimization or equilibrium)?
- How to solve an equilibrium problem?

Market clearing: **optimization vs equilibrium!**



Equilibrium

For each generator:

Maximize profit
subject to production limits

For each demand:

Maximize utility
subject to consumption limits

Price-setter's problem

Optimization

Maximize market's social welfare
subject to:

- Production limits of generators
- Consumption limits of demands
- Power balance

How to **solve** the equilibrium problem?

Equilibrium

For each generator:

Maximize profit

subject to production limits

For each demand:

Maximize utility

subject to consumption limits

Price-setter's problem

How to solve the equilibrium problem?

Equilibrium

For each generator:

Maximize profit

subject to production limits

For each demand:

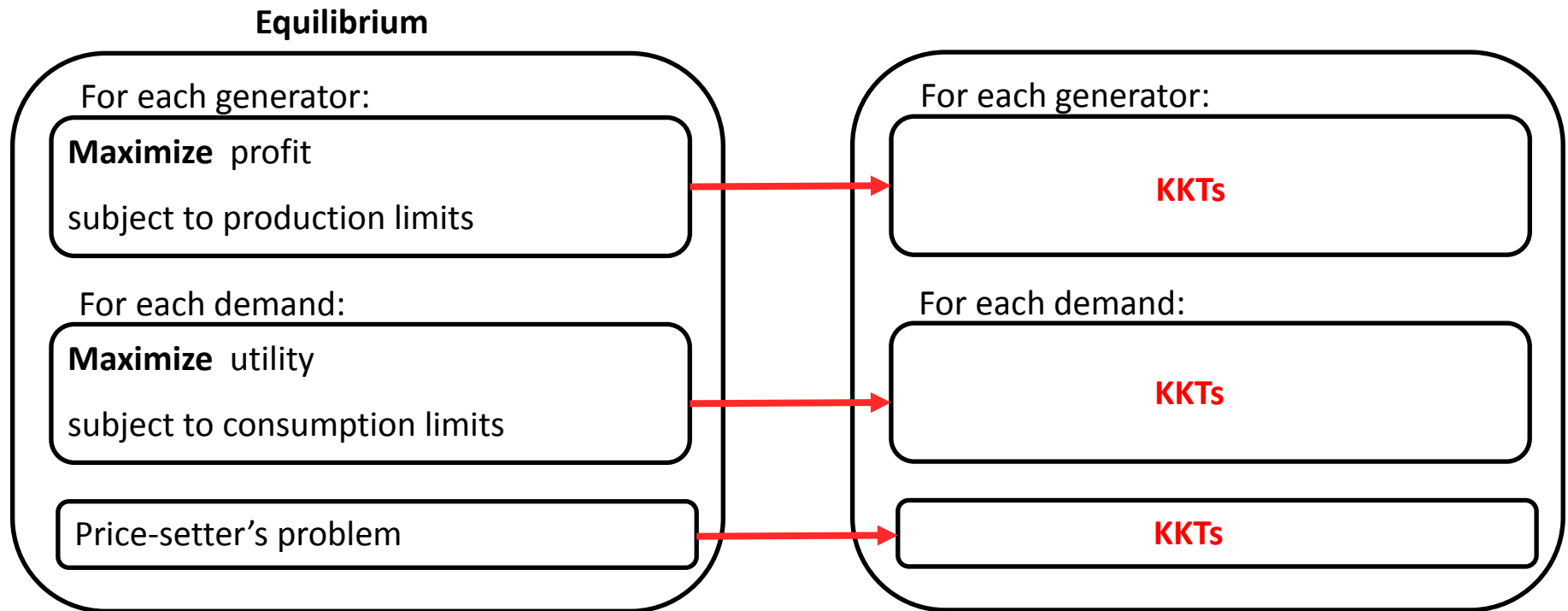
Maximize utility

subject to consumption limits

Price-setter's problem

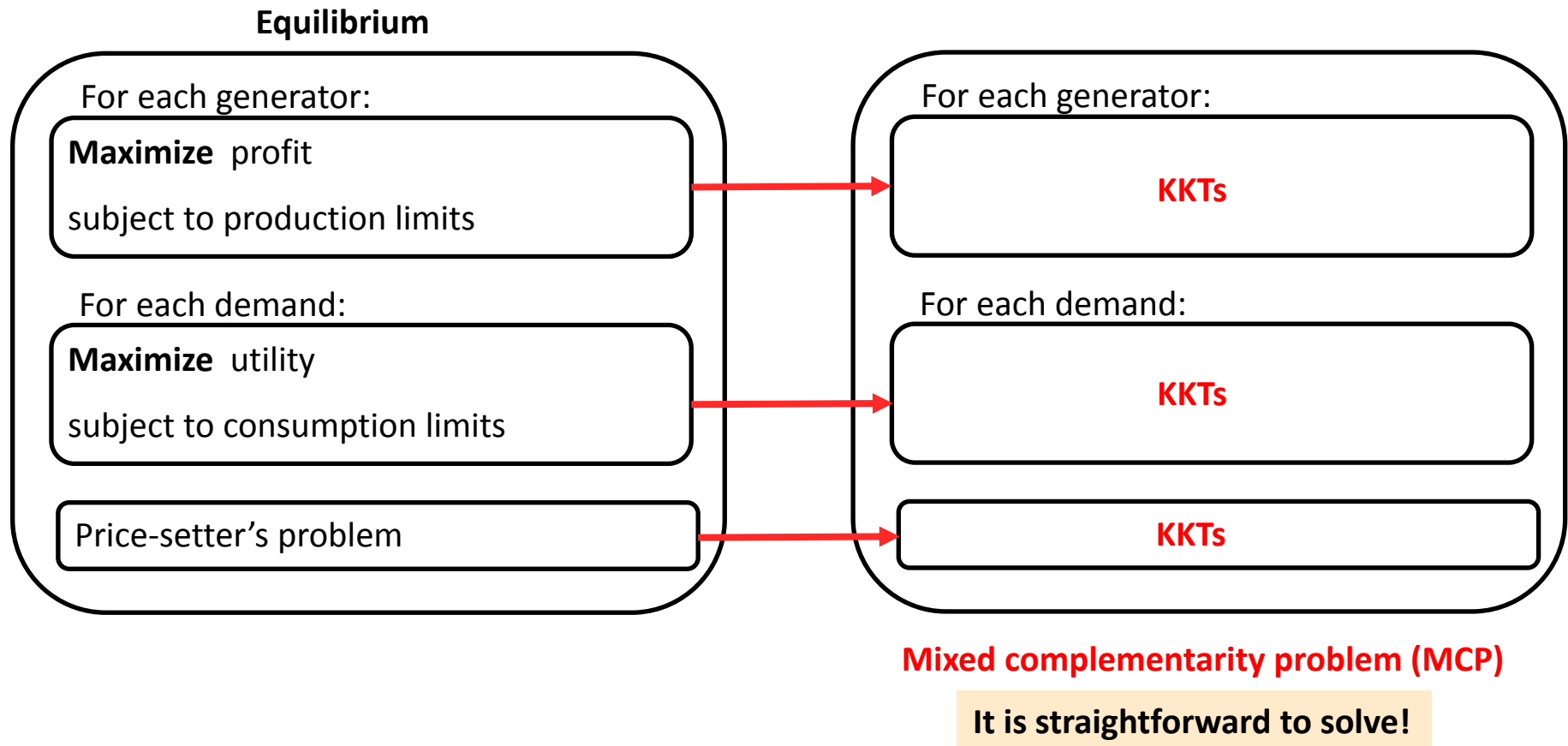
Replace each optimization problem within the equilibrium problem by its equivalent **Karush-Kuhn-Tucker (KKT)** optimality conditions! Recall that these conditions are a collection of equality and inequality conditions without any objective function!

How to solve the equilibrium problem?



Replace each optimization problem within the equilibrium problem by its equivalent **Karush-Kuhn-Tucker (KKT)** optimality conditions! Recall that these conditions are a collection of equality and inequality conditions without any objective function!

How to solve the equilibrium problem?



Market clearing: optimization vs equilibrium!



Equilibrium

For each generator:

Maximize profit

subject to production limits

For each demand:

Maximize utility

subject to consumption limits

Price-setter's problem

Optimization

Maximize market's social welfare

subject to:

- Production limits of generators
- Consumption limits of demands
- Power balance

Let's check again the equilibrium and optimization problems above!

Market clearing: optimization vs equilibrium!



Equilibrium

For each generator:

Maximize profit
subject to production limits

For each demand:

Maximize utility
subject to consumption limits

Price-setter's problem

MCP

Optimization

Maximize market's social welfare
subject to:

- Production limits of generators
- Consumption limits of demands
- Power balance

KKTs

Market clearing: optimization vs equilibrium!

Equilibrium

For each generator:

Maximize profit
subject to production limits

For each demand:

Maximize utility
subject to consumption limits

Price-setter's problem

Optimization

Maximize market's social welfare
subject to:

- Production limits of generators
- Consumption limits of demands
- Power balance

MCP

KKTs

Do the MCP (obtained from equilibrium) and the KKTs (obtained from optimization) include **identical** conditions?

If so, the equilibrium and optimization problems above are “**equivalent**”, i.e., any solution to the equilibrium problem is also a solution to the optimization problem and vice versa.

Market clearing as an equilibrium problem

Maximize $p^{G1}(\lambda - 12)$

subject to:

$$0 \leq p^{G1} \leq 100 \quad : \underline{\mu}^{G1}, \bar{\mu}^{G1}$$

Maximize $p^{G2}(\lambda - 20)$

subject to:

$$0 \leq p^{G2} \leq 80 \quad : \underline{\mu}^{G2}, \bar{\mu}^{G2}$$

Maximize $p^{D1}(40 - \lambda)$

subject to:

$$0 \leq p^{D1} \leq 100 \quad : \underline{\mu}^{D1}, \bar{\mu}^{D1}$$

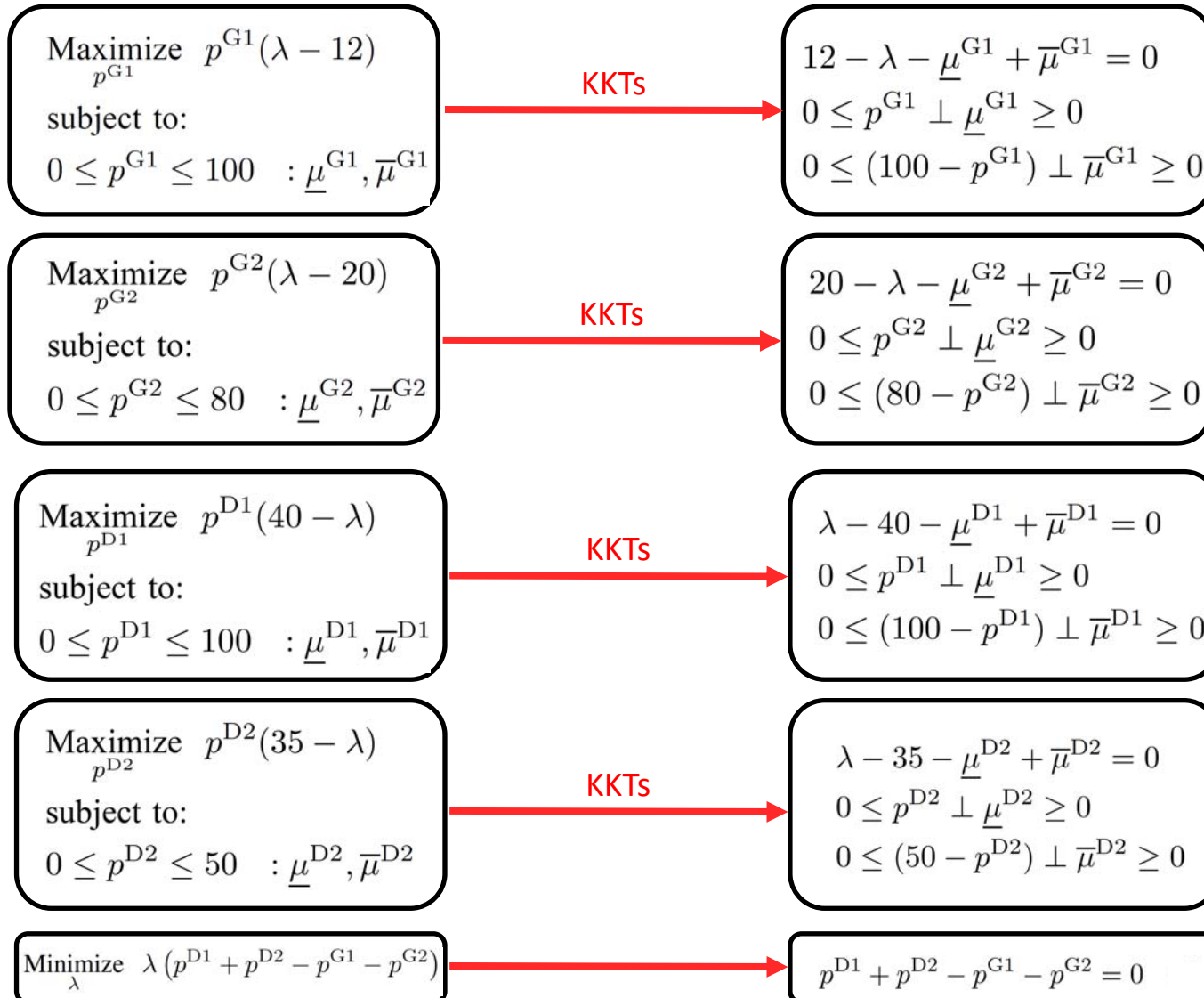
Maximize $p^{D2}(35 - \lambda)$

subject to:

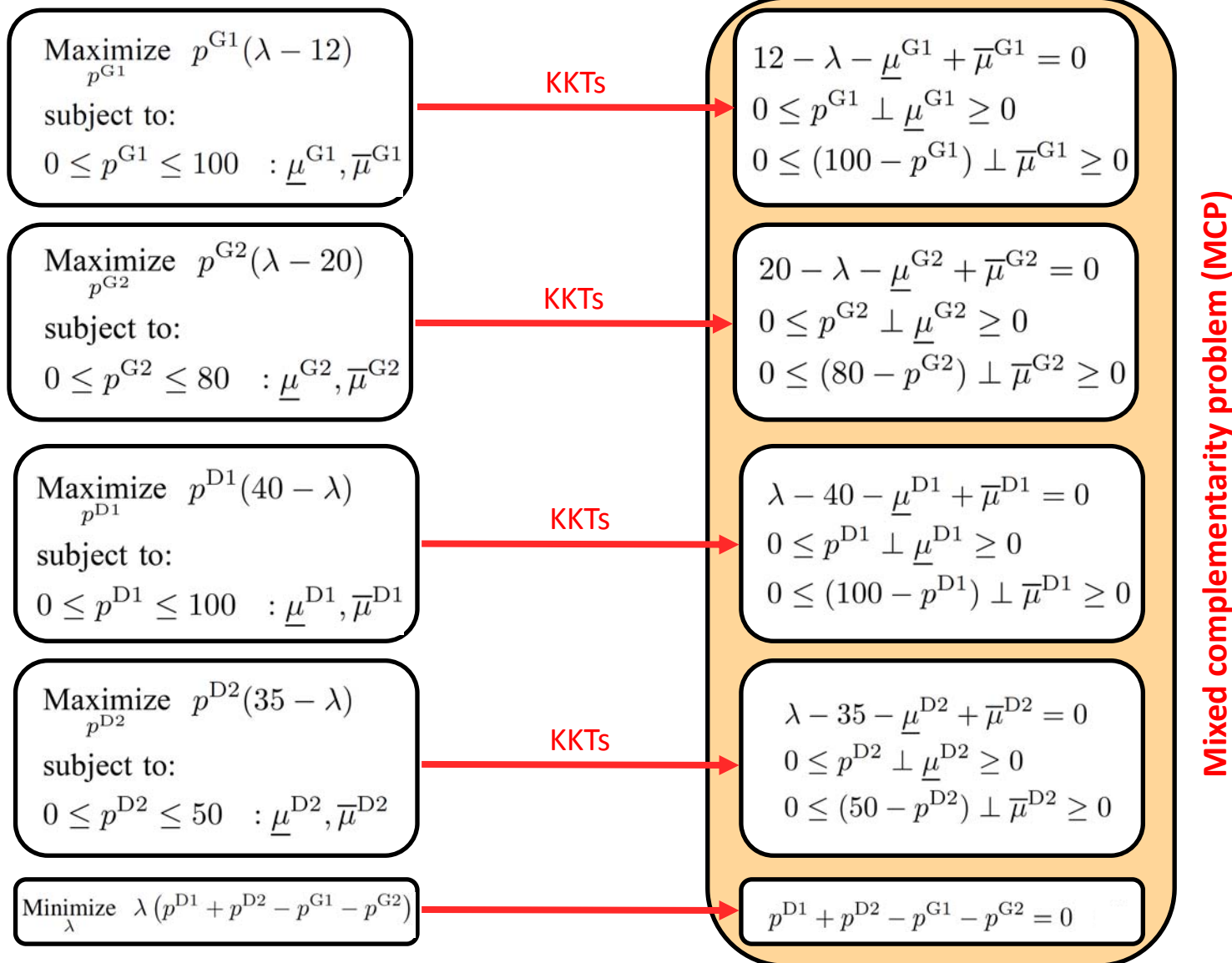
$$0 \leq p^{D2} \leq 50 \quad : \underline{\mu}^{D2}, \bar{\mu}^{D2}$$

Minimize $\lambda (p^{D1} + p^{D2} - p^{G1} - p^{G2})$

Market clearing as an equilibrium problem



Market clearing as an equilibrium problem



Market clearing as an equilibrium problem

$$12 - \lambda - \underline{\mu}^{G1} + \overline{\mu}^{G1} = 0$$

$$0 \leq p^{G1} \perp \underline{\mu}^{G1} \geq 0$$

$$0 \leq (100 - p^{G1}) \perp \overline{\mu}^{G1} \geq 0$$

$$20 - \lambda - \underline{\mu}^{G2} + \overline{\mu}^{G2} = 0$$

$$0 \leq p^{G2} \perp \underline{\mu}^{G2} \geq 0$$

$$0 \leq (80 - p^{G2}) \perp \overline{\mu}^{G2} \geq 0$$

$$\lambda - 40 - \underline{\mu}^{D1} + \overline{\mu}^{D1} = 0$$

$$0 \leq p^{D1} \perp \underline{\mu}^{D1} \geq 0$$

$$0 \leq (100 - p^{D1}) \perp \overline{\mu}^{D1} \geq 0$$

$$\lambda - 35 - \underline{\mu}^{D2} + \overline{\mu}^{D2} = 0$$

$$0 \leq p^{D2} \perp \underline{\mu}^{D2} \geq 0$$

$$0 \leq (50 - p^{D2}) \perp \overline{\mu}^{D2} \geq 0$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0$$

Mixed complementarity problem (MCP)

Market clearing as an equilibrium problem

Question:

Is this MCP identical to the KKTs of the market-clearing optimization problem?
Let's check!

$$12 - \lambda - \underline{\mu}^{G1} + \overline{\mu}^{G1} = 0$$

$$0 \leq p^{G1} \perp \underline{\mu}^{G1} \geq 0$$

$$0 \leq (100 - p^{G1}) \perp \overline{\mu}^{G1} \geq 0$$

$$20 - \lambda - \underline{\mu}^{G2} + \overline{\mu}^{G2} = 0$$

$$0 \leq p^{G2} \perp \underline{\mu}^{G2} \geq 0$$

$$0 \leq (80 - p^{G2}) \perp \overline{\mu}^{G2} \geq 0$$

$$\lambda - 40 - \underline{\mu}^{D1} + \overline{\mu}^{D1} = 0$$

$$0 \leq p^{D1} \perp \underline{\mu}^{D1} \geq 0$$

$$0 \leq (100 - p^{D1}) \perp \overline{\mu}^{D1} \geq 0$$

$$\lambda - 35 - \underline{\mu}^{D2} + \overline{\mu}^{D2} = 0$$

$$0 \leq p^{D2} \perp \underline{\mu}^{D2} \geq 0$$

$$0 \leq (50 - p^{D2}) \perp \overline{\mu}^{D2} \geq 0$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0$$

Mixed complementarity problem (MCP)

Market clearing as an equilibrium problem

Question:

Is this MCP identical to the KKTs of the market-clearing optimization problem?
Let's check!

Answer: ?

Optimization

$$\text{Maximize}_{p^{G1}, p^{G2}, p^{D1}, p^{D2}} SW = [40p^{D1} + 35p^{D2}] - [12p^{G1} + 20p^{G2}] \quad (1a)$$

subject to:

$$0 \leq p^{D1} \leq 100 : \underline{\mu}^{D1}, \bar{\mu}^{D1} \quad (1b)$$

$$0 \leq p^{D2} \leq 50 : \underline{\mu}^{D2}, \bar{\mu}^{D2} \quad (1c)$$

$$0 \leq p^{G1} \leq 100 : \underline{\mu}^{G1}, \bar{\mu}^{G1} \quad (1d)$$

$$0 \leq p^{G2} \leq 80 : \underline{\mu}^{G2}, \bar{\mu}^{G2} \quad (1e)$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0 : \lambda \quad (1f)$$

$$\begin{aligned} 12 - \lambda - \underline{\mu}^{G1} + \bar{\mu}^{G1} &= 0 \\ 0 \leq p^{G1} \perp \underline{\mu}^{G1} &\geq 0 \\ 0 \leq (100 - p^{G1}) \perp \bar{\mu}^{G1} &\geq 0 \end{aligned}$$

$$\begin{aligned} 20 - \lambda - \underline{\mu}^{G2} + \bar{\mu}^{G2} &= 0 \\ 0 \leq p^{G2} \perp \underline{\mu}^{G2} &\geq 0 \\ 0 \leq (80 - p^{G2}) \perp \bar{\mu}^{G2} &\geq 0 \end{aligned}$$

$$\begin{aligned} \lambda - 40 - \underline{\mu}^{D1} + \bar{\mu}^{D1} &= 0 \\ 0 \leq p^{D1} \perp \underline{\mu}^{D1} &\geq 0 \\ 0 \leq (100 - p^{D1}) \perp \bar{\mu}^{D1} &\geq 0 \end{aligned}$$

$$\begin{aligned} \lambda - 35 - \underline{\mu}^{D2} + \bar{\mu}^{D2} &= 0 \\ 0 \leq p^{D2} \perp \underline{\mu}^{D2} &\geq 0 \\ 0 \leq (50 - p^{D2}) \perp \bar{\mu}^{D2} &\geq 0 \end{aligned}$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0$$

Mixed complementarity problem (MCP)

Market clearing as an equilibrium problem

Question:

Is this MCP identical to the KKTs of the market-clearing optimization problem?
Let's check!

Answer: Yes!

Optimization

$$\text{Maximize}_{p^{G1}, p^{G2}, p^{D1}, p^{D2}} SW = [40p^{D1} + 35p^{D2}] - [12p^{G1} + 20p^{G2}] \quad (1a)$$

subject to:

$$0 \leq p^{D1} \leq 100 : \underline{\mu}^{D1}, \bar{\mu}^{D1} \quad (1b)$$

$$0 \leq p^{D2} \leq 50 : \underline{\mu}^{D2}, \bar{\mu}^{D2} \quad (1c)$$

$$0 \leq p^{G1} \leq 100 : \underline{\mu}^{G1}, \bar{\mu}^{G1} \quad (1d)$$

$$0 \leq p^{G2} \leq 80 : \underline{\mu}^{G2}, \bar{\mu}^{G2} \quad (1e)$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0 : \lambda \quad (1f)$$

KKTs

$$\begin{aligned} 12 - \lambda - \underline{\mu}^{G1} + \bar{\mu}^{G1} &= 0 \\ 0 \leq p^{G1} \perp \underline{\mu}^{G1} &\geq 0 \\ 0 \leq (100 - p^{G1}) \perp \bar{\mu}^{G1} &\geq 0 \end{aligned}$$

$$\begin{aligned} 20 - \lambda - \underline{\mu}^{G2} + \bar{\mu}^{G2} &= 0 \\ 0 \leq p^{G2} \perp \underline{\mu}^{G2} &\geq 0 \\ 0 \leq (80 - p^{G2}) \perp \bar{\mu}^{G2} &\geq 0 \end{aligned}$$

$$\begin{aligned} \lambda - 40 - \underline{\mu}^{D1} + \bar{\mu}^{D1} &= 0 \\ 0 \leq p^{D1} \perp \underline{\mu}^{D1} &\geq 0 \\ 0 \leq (100 - p^{D1}) \perp \bar{\mu}^{D1} &\geq 0 \end{aligned}$$

$$\begin{aligned} \lambda - 35 - \underline{\mu}^{D2} + \bar{\mu}^{D2} &= 0 \\ 0 \leq p^{D2} \perp \underline{\mu}^{D2} &\geq 0 \\ 0 \leq (50 - p^{D2}) \perp \bar{\mu}^{D2} &\geq 0 \end{aligned}$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0$$

Mixed complementarity problem (MCP)

Conclusions so far!

- The equilibrium and optimization forms of the market-clearing problem are **equivalent**, because their corresponding KKT conditions are identical!
- Both equilibrium and optimization forms of the market-clearing problem obtain the “**Nash equilibrium solution**”, i.e., no market player desires to deviate from the market-clearing outcomes!

Form 1: Market clearing as an optimization problem

$$\text{Maximize}_{p_g^G, p_d^D, \theta_n} \sum_d U_d p_d^D - \sum_g C_g p_g^G$$

subject to:

$$0 \leq p_d^D \leq \overline{P}_d^D : \underline{\mu}_d^D, \overline{\mu}_d^D \quad \forall d$$

$$0 \leq p_g^G \leq \overline{P}_g^G : \underline{\mu}_g^G, \overline{\mu}_g^G \quad \forall g$$

$$\sum_{d \in \Psi_n} p_d^D + \sum_{m \in \Omega_n} B_{n,m}(\theta_n - \theta_m) - \sum_{g \in \Psi_n} p_g^G = 0 \quad : \lambda_n \quad \forall n$$

$$-F_{n,m} \leq B_{n,m}(\theta_n - \theta_m) \leq F_{n,m} : \underline{\eta}_{n,m}, \overline{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

$$\theta_{(n=ref)} = 0 \quad : \gamma$$

Form 2: Market clearing as an **equilibrium** problem

Each generator:

$$\text{Maximize}_{p_g^G} p_g^G (\lambda_{n:g \in \Psi_n} - C_g)$$

subject to:

$$0 \leq p_g^G \leq \bar{P}_g^G : \underline{\mu}_g^G, \bar{\mu}_g^G$$

Each elastic demand:

$$\text{Maximize}_{p_d^D} p_d^D (U_d - \lambda_{n:d \in \Psi_n})$$

subject to:

$$0 \leq p_d^D \leq \bar{P}_d^D : \underline{\mu}_d^D, \bar{\mu}_d^D$$

Transmission owner as a spatial arbitrager (it buys power at a bus and sells it back at another one):

$$\text{Maximize}_{\theta_n} \sum_{n, (m \in \Omega_n)} \lambda_n [B_{n,m} (\theta_m - \theta_n)]$$

subject to:

$$-F_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq F_{n,m} : \underline{\eta}_{n,m}, \bar{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Price-setter:

$$\text{Minimize}_{\lambda_n} \sum_n \lambda_n \left(\sum_{d \in \Psi_n} p_d^D + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) - \sum_{g \in \Psi_n} p_g^G \right)$$

Form 2: Market clearing as an **equilibrium** problem

Each generator:

$$\text{Maximize}_{p_g^G} p_g^G (\lambda_{n:g \in \Psi_n} - C_g)$$

subject to:

$$0 \leq p_g^G \leq \bar{P}_g^G : \underline{\mu}_g^G, \bar{\mu}_g^G$$

Each elastic demand:

$$\text{Maximize}_{p_d^D} p_d^D (U_d - \lambda_{n:d \in \Psi_n})$$

subject to:

$$0 \leq p_d^D \leq \bar{P}_d^D : \underline{\mu}_d^D, \bar{\mu}_d^D$$

Transmission owner as a spatial arbitrager (it buys power at a bus and sells it back at another one):

$$\text{Maximize}_{\theta_n} \sum_{n, (m \in \Omega_n)} \lambda_n [B_{n,m} (\theta_m - \theta_n)]$$

subject to:

$$-F_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq F_{n,m} : \underline{\eta}_{n,m}, \bar{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

This term is known as "congestion rent"!

Price-setter:

$$\text{Minimize}_{\lambda_n} \sum_n \lambda_n \left(\sum_{d \in \Psi_n} p_d^D + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) - \sum_{g \in \Psi_n} p_g^G \right)$$

Closer look into congestion rent

Transmission owner as a spatial arbitrage (it buys power at a bus and sells it back at another one):

$$\text{Maximize}_{\theta_n} \sum_{n, (m \in \Omega_n)} \lambda_n [B_{n,m} (\theta_m - \theta_n)]$$

Let's investigate if this objective function is correct!

subject to:

$$-F_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq F_{n,m} : \underline{\eta}_{n,m}, \bar{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Closer look into congestion rent

Transmission owner as a spatial arbitrage (it buys power at a bus and sells it back at another one):

$$\text{Maximize}_{\theta_n} \sum_{n, (m \in \Omega_n)} \lambda_n [B_{n,m}(\theta_m - \theta_n)]$$

Question: What are “financial transmission rights (FTRs)”?

subject to:

$$-F_{n,m} \leq B_{n,m}(\theta_n - \theta_m) \leq F_{n,m} \quad : \underline{\eta}_{n,m}, \bar{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Form 2: Market clearing as an **equilibrium** problem

Each generator:

$$\text{Maximize}_{p_g^G} p_g^G (\lambda_{n:g \in \Psi_n} - C_g)$$

subject to:

$$0 \leq p_g^G \leq \bar{P}_g^G : \underline{\mu}_g^G, \bar{\mu}_g^G$$

Each elastic demand:

$$\text{Maximize}_{p_d^D} p_d^D (U_d - \lambda_{n:d \in \Psi_n})$$

subject to:

$$0 \leq p_d^D \leq \bar{P}_d^D : \underline{\mu}_d^D, \bar{\mu}_d^D$$

Transmission owner as a spatial arbitrager (it buys power at a bus and sells it back at another one):

$$\text{Maximize}_{\theta_n} \sum_{n, (m \in \Omega_n)} \lambda_n [B_{n,m} (\theta_m - \theta_n)]$$

subject to:

$$-F_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq F_{n,m} : \underline{\eta}_{n,m}, \bar{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Price-setter:

$$\text{Minimize}_{\lambda_n} \sum_n \lambda_n \left(\sum_{d \in \Psi_n} p_d^D + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) - \sum_{g \in \Psi_n} p_g^G \right)$$

Form 3: Market clearing as an MCP

$$-U_d + \lambda_{n:d \in \Psi_n} - \underline{\mu}_d^D + \bar{\mu}_d^D = 0 \quad \forall d$$

$$C_g - \lambda_{n:g \in \Psi_n} - \underline{\mu}_g^G + \bar{\mu}_g^G = 0 \quad \forall g$$

$$\sum_{m \in \Omega_n} B_{n,m} (\lambda_n - \lambda_m + \bar{\eta}_{n,m} - \bar{\eta}_{m,n} - \underline{\eta}_{n,m} + \underline{\eta}_{m,n}) + (\gamma)_{n=ref} = 0 \quad \forall n$$

$$\sum_{d \in \Psi_n} p_d^D + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) - \sum_{g \in \Psi_n} p_g^G = 0 \quad \forall n$$

$$\theta_{(n=ref)} = 0$$

$$0 \leq p_g^G \perp \underline{\mu}_g^G \geq 0 \quad \forall g$$

$$0 \leq [\bar{P}_g^G - p_g^G] \perp \bar{\mu}_g^G \geq 0 \quad \forall g$$

$$0 \leq p_d^D \perp \underline{\mu}_d^D \geq 0 \quad \forall d$$

$$0 \leq [\bar{P}_d^D - p_d^D] \perp \bar{\mu}_d^D \geq 0 \quad \forall d$$

$$0 \leq [F_{n,m} + B_{n,m} (\theta_n - \theta_m)] \perp \underline{\eta}_{n,m} \geq 0 \quad \forall n, \forall m \in \Omega_n$$

$$0 \leq [F_{n,m} - B_{n,m} (\theta_n - \theta_m)] \perp \bar{\eta}_{n,m} \geq 0 \quad \forall n, \forall m \in \Omega_n$$

Conclusion

All three forms of the market-clearing problem, i.e.,

- optimization
- equilibrium
- MCP

are equivalent!

Exercises

1- Which form of the market-clearing problem (optimization or equilibrium) is more appealing to market operators?

2- Consider an equilibrium form of the market-clearing problem. Is it possible to solve it iteratively without deriving KKTs? If so, what are the pros and cons?

Guide: Consider an iterative mechanism, in which the market operator fixes a set of initial prices, and then each market participant makes her own dispatch decisions accordingly. Based on the participants' dispatch decisions, the market operator checks whether nodal power balance conditions hold or not. If not, the operator "systematically" adjusts those prices and disseminates the updated prices among participants. This can be continued until there is no demand-supply mismatch. If interested, read about "Walrasian auction" and its "tâtonnement process", which indeed requires a decomposition technique (e.g., Lagrangian relaxation or ADMM).

3- For a given Nash equilibrium (NE) problem, how to mathematically identify that an equivalent optimization problem exists?

Guide: check chapter 4 of the book by S. Gabriel et al. (available on DTU Inside) and learn about "Principle of Symmetry", referring to the symmetry of Jacobian matrix. Search how we can derive the Jacobian matrix of a game. You can also check Theorem 1.3.1 of the book by F. Facchinei and J.-S. Pang (available on DTU Inside).

4- Investigate how the solution existence and the solution uniqueness for a Nash equilibrium (NE) can be mathematically proven.

Guide: For uniqueness, search about "monotonicity" property of a game. How can we ensure a game is strongly monotone by checking the Jacobian matrix? You can also read about "degree theory".

GAMS code for MCP to be solved by PATH

Sets

```
g  generators /G1*G2/
n  buses      /N1*N3/
d  demands    /D1*D2/
alias (n,m)
```

sets

```
MapN(n,n) Network topology /
N1.N2
N1.N3
N2.N3
N2.N1
N3.N1
N3.N2/
```

```
MapG(g,n) Location of generators /
G1.N1
G2.N2
/
```

```
MapD(d,n) Location of demands /
D1.N2
D2.N3
/
;
```

GAMS code for MCP to be solved by PATH

```
Parameter PGmax(g) Capacity of generators [MW] /
```

```
G1 100
```

```
G2 80
```

```
/;
```

```
Parameter C(g) Offer price of generators [$ per MWh] /
```

```
G1 12
```

```
G2 20
```

```
/;
```

```
Parameter L(d) Maximum load of demands [MW] /
```

```
D1 100
```

```
D2 50
```

```
/;
```

```
Parameter U(d) Utility of demands [$ per MWh] /
```

```
D1 40
```

```
D2 35
```

```
/
```

```
;
```

GAMS code for MCP to be solved by PATH

Table Fmax(n,n) Capacity of transmission lines [MW]

	N1	N2	N3
N1	0	100	100
N2	100	0	100
N3	100	100	0

;

Table B(n,n) Susceptance of transmission lines [Ohm⁻¹]

	N1	N2	N3
N1	0	500	500
N2	500	0	500
N3	500	500	0

;

GAMS code for MCP to be solved by PATH

Free variable

```
p_D(d)      Consumption level of demand d [MW]
p_G(g)      Production level of generator g [MW]
theta(n)    Voltage angle of bus n [rad]
lambda(n)   Dual var.: locational marginal price [$ per MWh]
gamma       Dual var. associated with equality constraint introducing ref. bus
;
```

Positive variable

```
mu_D_min(d) Dual var. associated with lower bound of consumption level
mu_D_max(d) Dual var. associated with upper bound of consumption level
mu_G_min(g) Dual var. associated with lower bound of production level
mu_G_max(g) Dual var. associated with upper bound of production level
eta_min(n,m) Dual var. associated with transmission capacity constraints
eta_max(n,m) Dual var. associated with transmission capacity constraints
;
```

GAMS code for MCP to be solved by PATH

Equations

```
cons1,cons2,cons3,cons4,cons5,cons6,cons7,cons8,cons9,cons10,cons11,cons12;
```

* Primal constraints

```
cons1(g) ..      p_G(g) =g= 0;
cons2(g) ..      -p_G(g) =g= -PGmax(g);
cons3(d) ..      p_D(d) =g= 0;
cons4(d) ..      -p_D(d) =g= -L(d);
cons5(n,m) ..    [B(n,m)*(theta(n)-theta(m))] =g= -Fmax(n,m);
cons6(n,m) ..    -[B(n,m)*(theta(n)-theta(m))] =g= -Fmax(n,m);
cons7 ..         theta('N1') =e= 0;
cons8(n) ..      -sum(g$MapG(g,n),p_G(g))+sum(d$MapD(d,n),p_D(d))
                  +sum(m$MapN(n,m),B(n,m)*(theta(n)-theta(m))) =e= 0;
```

* KKT conditions

```
cons9(d) ..      -U(d)+sum(n$MapD(d,n),lambda(n))+mu_D_max(d)-mu_D_min(d) =e= 0;
cons10(g) ..     C(g)-sum(n$MapG(g,n),lambda(n))+mu_G_max(g)-mu_G_min(g) =e= 0;
cons11(n)$ (ord(n) eq 1) .. sum(m$MapN(n,m),B(n,m)*[lambda(n)-lambda(m)
                  +eta_max(n,m)-eta_max(m,n)-eta_min(n,m)
                  +eta_min(m,n)]) + gamma =e= 0;
cons12(n)$ (ord(n) <> 1) .. sum(m$MapN(n,m),B(n,m)*[lambda(n)-lambda(m)
                  +eta_max(n,m)-eta_max(m,n)-eta_min(n,m)
                  +eta_min(m,n)]) =e= 0;
```

GAMS code for MCP to be solved by PATH

```
Model Market_clearing /  
  cons1.mu_G_min  
  cons2.mu_G_max  
  cons3.mu_D_min  
  cons4.mu_D_max  
  cons5.eta_min  
  cons6.eta_max  
  cons7.gamma  
  cons8.lambda  
  cons9  
  cons10  
  cons11  
  cons12/;  
  
Solve Market_clearing using mcp;  
option mcp = PATH;  
Display p_G.l,p_D.l,lambda.l;
```

Thanks for your attention!

Email: seykaz@elektro.dtu.dk